Tomato bacterial and Viral diseases

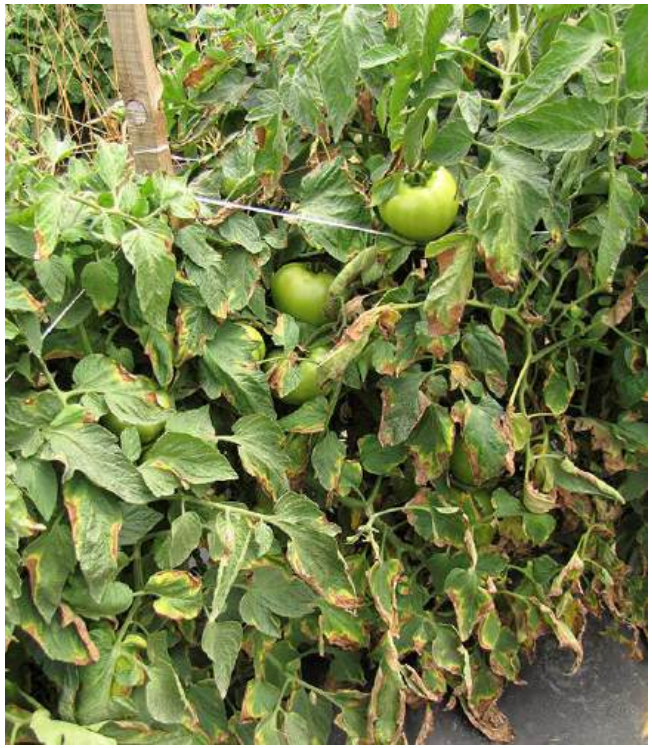
Bacterial diseases

1. Bacterial canker - C*lavibacter michiganensis* pv. *michiganensis*
2. Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*
3. Bacterial wilt- *Ralstonia solanacearum*

Viral diseases

1. Spotted wilt - Tomato Spotted Wilt Virus- (TSWV)
2. Mosaic - *Tomato Mosaic Virus* -(ToMV)
3. Leaf curl -*Tomato Yellow Leaf Curl Virus* (TYLCV)

Bacterial canker - *Clavibacter michiganensis* pv. *michiganensis*



Bacterial canker - *Clavibacter michiganensis* pv. *michiganensis*

Bacteria in veins ooze out when the veins are cut.

Bacterial streaming can be seen from cut by placing a piece of leaf tissue in water on a microscope slide

Bacterial streaming is diagnostic for canker distinguishing it from drought stress.

Bacterial canker - *Clavibacter michiganensis* pv. *michiganensis*

Bacterial canker - *Clavibacter michiganensis* pv. *michiganensi*

- The images are the upper and lower surfaces of the same leaflets.
- A less common symptom is black veins on the underside of leaves.
- Lots of bacteria streamed from symptomatic tissue including the black veins.

Bacterial canker - *Clavibacter michiganensis* pv. *michiganensis*

Bacterial canker - *Clavibacter michiganensis* pv. *michiganensis*

Bacterial canker - *Clavibacter michiganensis* pv. *michiganensis*

Symptoms:

- Seedlings**:** Marginal necrosis, tan to dark necrotic patches on the leaves and stems, and small white raised blisters on infected leaves may be symptoms of bacterial canker infection on young plants.
- Stunting, wilting and stem splitting can also occur, especially in grafted seedlings.
- However, **symptoms development takes many weeks after infection and therefore may not be visible** at the seedling or transplant stage.

Bacterial canker - C*lavibacter michiganensis* pv. *michiganensis*

Symptoms:

Leaf and plant:

- Leaf yellowing and necrosis along leaf margins called as firing or marginal necrosis can indicate a foliar and/or systemic infection.
- When the stems or petioles are cut open, discoloration of the vascular tissues may be seen.
- In greenhouse grown plants, symptoms appear as interveinal chlorotic to pale green patches that quickly become necrotic, giving a scorched appearance.
- Infected plants wilt, starts with the lower older leaves, or leaves above the point of infection.

Bacterial canker - C*lavibacter michiganensis* pv. *michiganensis*

Symptoms contd…

- Wilting may be asymmetric, appearing more on one side of the plant than the other.
- Typical leaf symptoms are brown tissue along the margin with a yellow border (marginal burning) and occasional v-shaped pattern of the affected tissue.
- Curling of leaves and wilting occurs as the disease progresses because the bacteria clog up veins.
- Sometimes there is no yellow border, which may reflect variety or environmental conditions

Bacterial canker - *Clavibacter michiganensis* pv. *michiganensis*

Symptoms contd…

- Infected leaves die, **and light brown streaks or cankers develop on infected stems,** which may darken with age.
- Typical cankers can be common in the field but are rarely seen in the greenhouse.
- The **vascular tissues become light brown to reddish brown and the pith appears mealy, brown and dry.**
- Older plants tend to be less susceptible than younger ones and the disease tends to be more severe on plants if infected in early stages that in the later stage.

Bacterial canker - *Clavibacter michiganensis* pv. *michiganensis*

Symptoms:

- **Fruits**: Small light brown spots that are often raised and may have a white border.
- Small dark spots surrounded by a white halo or bird‘s-eye are characteristic of bacterial canker on field-grown fruit.
- Spots become raised and the centers turn brown with age.
- Infected fruits, if cut open, internal yellowing or browning due to decay of the tissues may be seen.
- **In the greenhouse, bird‘s-eye spots are typically not observed, but fruit may appear netted or marbled, or they may remain symptomless**

Bacterial canker - C*lavibacter michiganensis* pv. *michiganensis*

Pathogen:

Disease is caused by *Clavibacter michiganensis* sub sp. *michiganensis* (Cmm)

The bacterium is aerobic, Gram + ve,
coryneform rod with cell growing either singly or in pairs.

Bacterial canker - C*lavibacter michiganensis* pv. *michiganensis*

Disease cycle and epidemiology

- The bacterium perennates in infected seed, plant debris and weed hosts.
- Optimum temperature for disease development is 28°C.
- The disease is more severe in wet weather.

Bacterial canker - *Clavibacter michiganensis* pv. *michiganensis*

Management

- Collect and destroy all infected plant debris.
- Follow crop rotation and tomato should not be grown in infested field for at least 3 years.
- **Extraction of seeds by anaerobic fermentation**
- Use disease free seed and treat them by dipping in the Streptocycline (100 ppm) solution for 1 h.
- Spray the crop with Streptocycline (100 ppm) followed by copper oxychloride (0.30%) or Bordeaux mixture (4:4:50) and repeat at 7-10 days interval.

Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

On older leaves as water-soaked areas , turn from yellow or light green to black or dark brown

Older spots are black, slightly raised, superficial and measure up to 0.3 inch (7.5 mm) in diameter and are surrounded by yellow halos

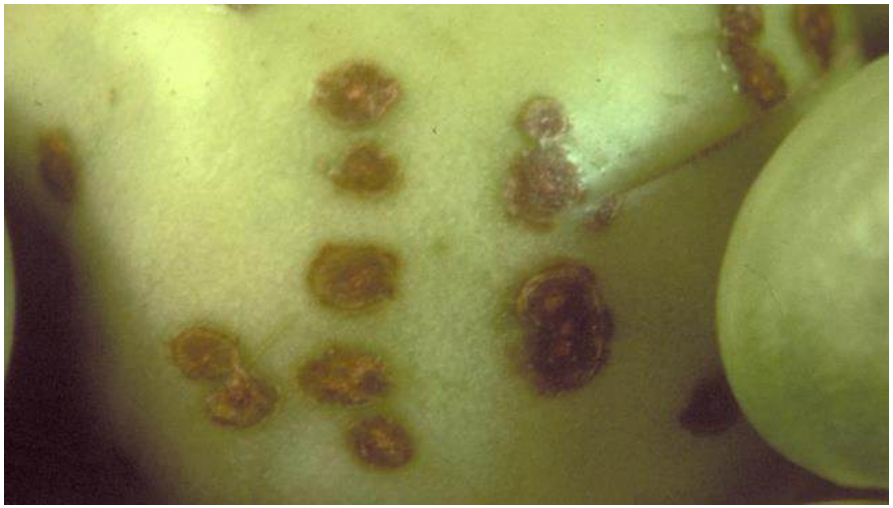
Larger leaf blotches may also occur, especially on the margins of leaves.

Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

Figure 1: Bacterial spot lesions on tomato leaves.

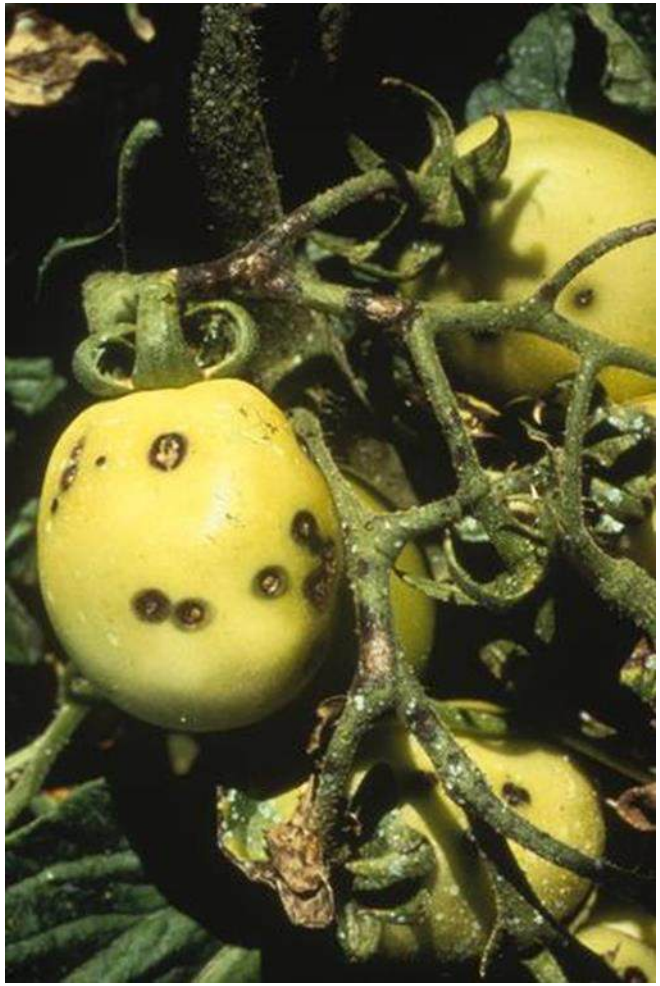
Photo 2. Oval spots on the leaf stalks, bacterial leaf spot, *Xanthomonas vesicatoria.*

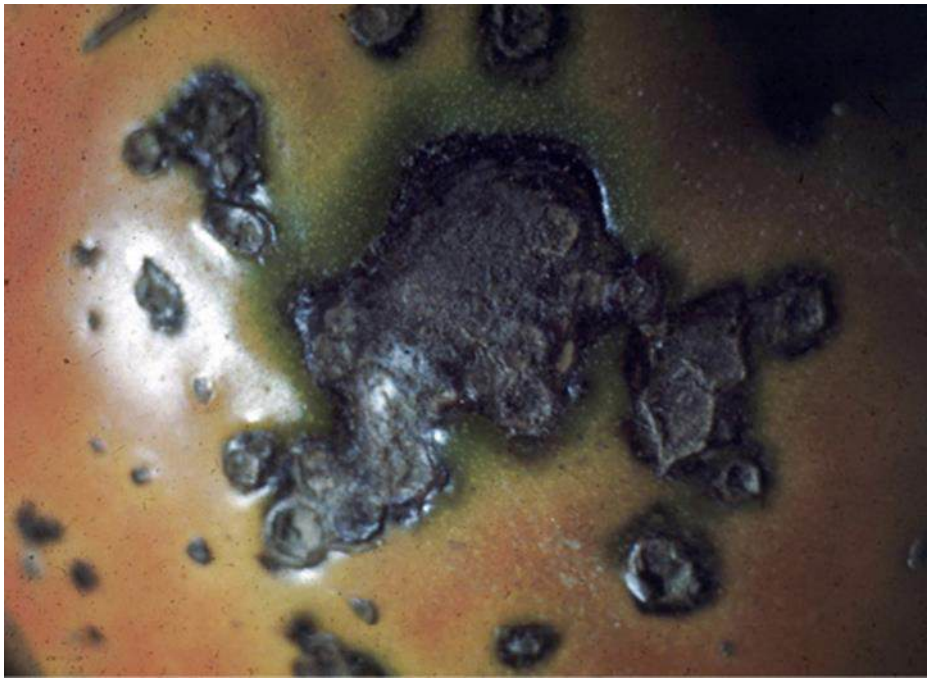
Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*



Lightly sunken and surrounded by a water-soaked halo, which soon disappears.

Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*





Scabby, circular, brown lesions on these immature fruit spot

Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

Crops: Tomato, capsicum, chilli and Other hosts: Some solanaceous weeds, especially nightshades

Symptoms and Signs:

- On leaves, small, irregular areas with a grassy appearance develop first.
- These areas dry out and form slightly raised dry spots that are greyish brown, particularly at the centre.
- The bacteria often ooze from these spots and, when dry, form a glistening, cream-coloured film on and around the lesions.
- Where infection is severe, the dried-out parts may combine and kill large areas of leaf. Marginal and tip burns on leaves have often been noted.

Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

Symptoms and Signs:

- On flowers, water-soaked dark-brown-to-black areas develop, which later dry out and turn grey.
- Flower infection often causes blossoms and young fruit to wither and fall.
- Main stems are occasionally attacked.
- Spots are often elongated but retain the same greyish, scab-like characteristics as other affected parts.
- Small, water-soaked areas develop on green fruit. These dry out and form slightly raised and wrinkled, brown-to-grey, scab-like bodies, making the fruit unmarketable and susceptible to secondary rots.

Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

Sources of infection: The disease is usually introduced first on the seed.

Spread by: Infected seed and water droplets

The bacteria, carried in water droplets, can form new spots on the leaves, stems or fruit where the droplet comes to rest.

In wet weather, particularly if strong winds are blowing, the disease may spread rapidly through a crop from a few affected plants.

Overhead irrigation acts in a similar manner to rain. Once the bacterium is established in the soil it may persist for 2 or 3 yrs.

Favoured by: Wet, windy weather, temperature of 24-30 ºC.

Tomato bacterial wilt- *Ralstonia solanacearum*

Tomato bacterial wilt- *Ralstonia solanacearum*

Tomato bacterial wilt- *Ralstonia solanacearum*

Tomato bacterial wilt- *Ralstonia solanacearum*

- Bacterial wilt is one of the most devastating disease in solanaceous plants.
- It is more serious in warm temperate, subtropical and tropical regions of the world.
- The disease is also called southern bacterial wilt, solanaceous wilt, southern bacterial blight and by many other common names in countries wherever it occurs.

Symptoms

- Characteristic symptoms of bacterial wilt on most of the hosts are sudden wilting and death of infected plants.
- Petioles of the older leaves of the infected plants droop down without yellowing.

Tomato bacterial wilt- *Ralstonia solanacearum*

Symptoms

- Leaves show epinasty symptoms accompanied by yellowing and stunting of whole plant.
- There is a typical browning of the vascular tissues of roots and stems
- Bacterial polysaccharides mechanically block the vascular system, which checks the translocation of water and minerals resulting in wilting of plants.
- Excessive development of adventitious roots may also take place in tomato.
- Cross sectioned part of the affected root or stem yield whitish milky bacterial ooze.

Tomato bacterial wilt- *Ralstonia solanacearum*

Pathogen:

- The disease is caused by *Ralstonia solanacearum*
- The bacterium is gram negative, rod shaped, frequently occurs in pairs, motile with 1 to 4 polar flagella and measures 0.5-0.7 x 1.5 –2.5 µm.
- Five races (race1, 2, 3, 4, 5) and four biovars (biovars I, II, III and IV) are known to occur in different parts of the world.
- In India race 1 and 3 and biovar II, III and IV are prevalent.

Tomato bacterial wilt- *Ralstonia solanacearum*

Disease cycle and epidemiology:

- The pathogen is both soil and seed borne in nature and overwinters in infected plant parts, in potato tubers, in wild host plants and solanaceous weeds and can survive at least for 2 years in the absence of any host.
- Injured and decaying infected tissues release bacteria in the soil which enter plants through wounds occurring in roots due to cultural practices, nematodes and insects.
- Relatively high soil moisture and soil temperature favour the disease.

Tomato bacterial wilt- *Ralstonia solanacearum*

Management

- Control of bacterial wilt in infested soil is very difficult as all the commercial cultivars are susceptible and no chemical control is available.
- Follow long crop rotation with non-solanaceous crops.
- Avoid the movement of water from infected plant to healthy plants.
- Shifting of date of transplanting to avoid period of high temperature, heavy rainfall or both.
- Green manuring or biofumigation with *Brassica* spp. may reduce the bacterial inoculum in soil.
- Bacterial bioagents - *Pseudomonas fluorescens*, *P. glumae*, *P. cepacia* and *Bacillus* spp.

Tomato bacterial wilt- *Ralstonia solanacearum*

Management

- Application of bleaching powder (15 kg/ ha) has also been found effective.
- The disease can also be controlled effectively if dazomet application is combined with soil solarization.
- Seedling dip in Streptocycline (100 ppm) for 30 minutes is also effective to some extent.
- Use resistant cvs./hybrids for cultivation

  Varieties - Arka Abha, Arka Alok

  F1 hybrids - Arka Abhed, Arka Samrat, Arka Rakshak, Arka Aditya

Tomato leaf curl -*Tomato Yellow Leaf Curl Virus* (TYLCV)

Tomato leaf curl - *Tomato Yellow Leaf Curl Virus* (TYLCV)

Upward curling and cupping

Curling and rolling

Yellowing and curling

Tomato leaf curl - *Tomato Yellow Leaf Curl Virus* (TYLCV)

Symptoms:

- The major symptoms include chlorosis of leaflets and reduction in their size accompanied by curling inwards.
- Significant reduction of nodes and internodal length occurs giving the plant a bushy appearance.
- In advanced stages of infection severe stunting and partial to complete sterility occurs.
- Infected plants bear few or no fruits.

Tomato leaf curl - *Tomato Yellow Leaf Curl Virus* (TYLCV)

Pathogen:

- The disease is caused by ***Tomato Yellow Leaf Curl Virus*** (TYLCV) which belongs to the Gemini Virus group.
- The particles are geminate, non-enveloped, 18 nm in diameter and genome consists of single stranded DNA.

Tomato leaf curl - *Tomato Yellow Leaf Curl Virus* (TYLCV)

Disease cycle and epidemiology:

- The virus **is transmitted through grafting** but not through sap, contacts between plants or seeds.
- The sole agency of its transmission is the **white fly** ***Bemisia tabaci*** that transmits the virus in a persistent manner.
- The virus has a wide host range including cultivated species, weed hosts and ornamentals.

Tomato Spotted Wilt -*Tomato Spotted Wilt Virus* (TSWV)

Another characteristic is **netting pattern** on leaves, and in some cases many leaves.

Leaf spots can entirely cover whole leaves and many leaves throughout the plant. At this stage the symptom is pronounced and very characteristic of tomato spotted wilt.

Tomato Spotted Wilt -*Tomato Spotted Wilt Virus* (TSWV)

A characteristic symptom of plants affected by tomato spotted wilt is leaf spots which form **concentric patterns.**

Leaf spots may resemble that of bacterial spot. Leaf spots associated with tomato spotted wilt have a **unique blotched lesion appearance** compared to bacterial spot that tends to break in the center.

Tomato Spotted Wilt -*Tomato Spotted Wilt Virus* (TSWV)

- **Streaking on the petioles and stems.**
- **Bronzing of leaves, which turn brown and wither**

Tomato Spotted Wilt -*Tomato Spotted Wilt Virus* (TSWV)

Tomato Spotted Wilt -*Tomato Spotted Wilt Virus* (TSWV)

- Death of the **terminal bud.**

Tomato Spotted Wilt -*Tomato Spotted Wilt Virus* (TSWV)

Leaf purpling and leaf upward curling is another symptom of plants affected with tomato spotted wilt. However, other diseases can also cause similar symptoms.

Common to see plants affected with tomato spotted wilt having high number of adventitious root formation.

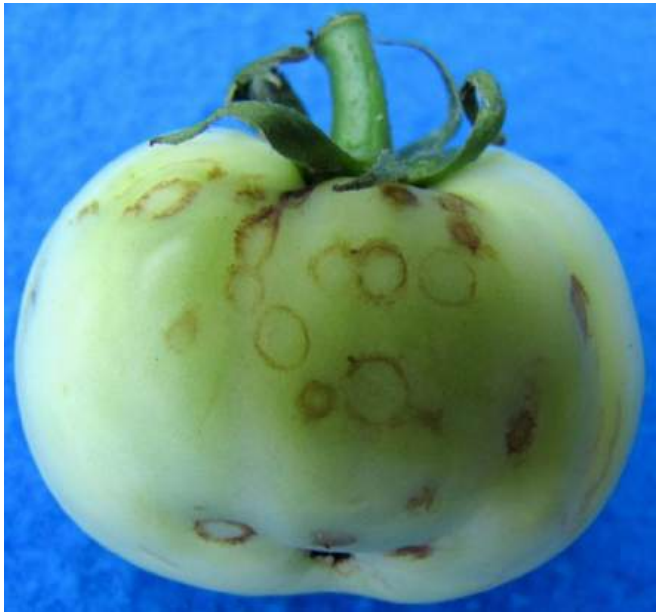
Other diseases and abiotic stress can also cause similar symptoms.

Tomato Spotted Wilt -*Tomato Spotted Wilt Virus* (TSWV)

Severe wilting may have symptoms of **severe necrosis and collapse of the pith.**

Severe stunting of the plants very early after transplanting into the field.

Tomato Spotted Wilt -*Tomato Spotted Wilt Virus* (TSWV)



Tomato Spotted Wilt -*Tomato Spotted Wilt Virus* (TSWV)

Symptoms

- Symptoms vary among hosts and in a single host species
- Stunting is a common symptom of TSWV infection
- Chlorotic or necrotic rings form on the leaves of many infected hosts
- Pale red or yellow areas with concentric circular marking in the normal red skin of ripe tomato are formed
- Discoloration of seed
- Thickening of veins and bronzing of young leaves

Tomato Spotted Wilt -*Tomato Spotted Wilt Virus* (TSWV)

Symptoms contd…

- Growing tips may die-back and terminal branches may be streaked
- Affected plants may have a one sided growth habit or may be entirely stunted and have drooping leaves, suggesting a wilt

Tomato Spotted Wilt -*Tomato Spotted Wilt Virus* (TSWV)

Pathogen:

- The disease is caused by *Tomato Spotted Wilt Virus* (TSWV), Tospo virus.
- The virus particles are isometric, enveloped, and 85 nm in diameter.
- The genome consists of single stranded RNA.

Tomato Spotted Wilt -*Tomato Spotted Wilt Virus* (TSWV)

Disease cycle and epidemiology:

- The virus is transmitted by thrips in persistent manner.
- The virus is acquired only by intensive feeding of the larvae and transmitted by adults.
- The virus has wide host range including bell pepper, lettuce, pea, tobacco, potato and large number of ornamental plant species.

Tomato Spotted Wilt -*Tomato Spotted Wilt Virus* (TSWV)

Management

- Removal of weed hosts
- Sow **5-6 rows of barrier crops** like maize, Jowar or Bajara 2 months before transplanting seedlings around the structure
- Prevent the entry of vectors
- Remove infected plants.
- Foliar sprays of either Dimethoate (0.05%) or Monocrotophos (0.05%) or Metasystox (0.02%) at 10 days intervals.
- After fruit formation spray, krishi oil (2%) at 10 days intervals.

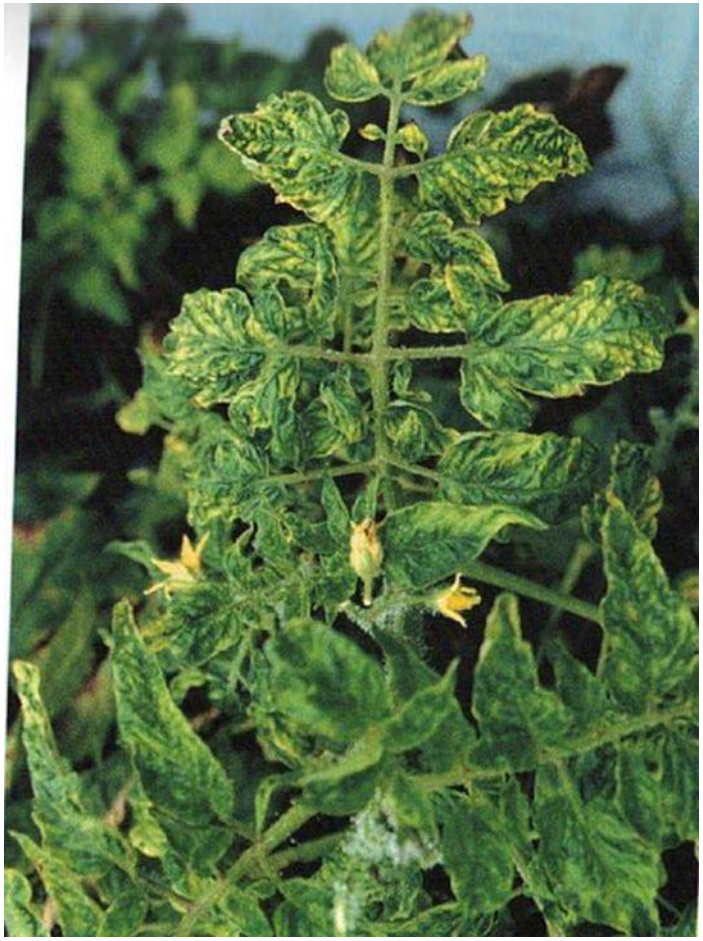
Tomato Mosaic Virus -(ToMV)

- **Light and dark green mottling on the leaves**
- **Distorted, puckered and smaller leaves**

- **Stunted and pale green than normal plant**

Tomato Mosaic Virus -(ToMV)

Tomato Mosaic Virus -(ToMV)



Tomato Mosaic Virus -(ToMV)

Symptoms:

- Mosaic is characterized by presence of dark and light green patches on the leaflets which may get distorted and puckered

- Malformation and reduction in leaf size may also occur.
- Mottling is best viewed by partially shading the leaf

- In case of early infection the plants are reduced in size and remain stunted.

- Reduced fruit set and uneven fruit ripening

Tomato Mosaic Virus -(ToMV)

Pathogen:

- *Tomato mosaic virus* (ToMV) is a member of Tobamo virus group.

- ToMV is an RNA-containing virus with straight rod particles about 300 x 18 nm.

Tomato Mosaic Virus -(ToMV)

Disease cycle and Epidemiology:

- Virus transmission in the field either occurs through contact between plants and field implements.
- The other modes of transmission are through sap and grafting but there is no vector involved.
- The virus is externally seed borne.
- The infection occurs during transplanting and the virus has a wide host range.

Tomato viral diseases

Management:

- The viral diseases of tomato can be managed effectively through use of virus free seed (ToMV).
- Cultivation of tomato leaf curl resistant **F1 hybrids - Arka Abhed, Arka Samrat, Arka Rakshak, Arka Aditya**
- Remove alternate hosts (TYLCV, TSWV).
- Use insecticides against the vectors and cultivation of resistant varieties / hybrids.

Tomato viral diseases

Management:

- Seedling bed should be in isolation from ornamental plants and susceptible crops and the surrounding areas should be kept free from weeds.
- Fine mesh netting may be useful for excluding thrips and other insect vectors.
- Aqueous extracts of neem (3-5ml/litre water) are known to possess strong antiviral substances against ToMV.

Cucumber Mosaic Virus